

Question	Marking details	Marks available					
		AO1	AO2	AO3	Total	Maths	Prac
6	Indicative content - Pyrophosphatase has tertiary structure - The active site has a specific shape - Pyrophosphate has a complementary shape - fits into the active site/ forms Enzyme Substrate Complex - lock and key theory/ induced fit - Phenylalanine must be a non-competitive inhibitor; - As it has a shape different to pyrophosphate; - Phenylalanine bonds to the allosteric site/site away from the active site; - Causes a change in the shape of the active site; - Pyrophosphate's active site is no longer complementary/prevents pyrophosphate binding - fewer enzyme-substrate complexes are formed; - The maximum rate of reaction cannot be reached at any concentration of pyrophosphate; - Phosphate must be a competitive inhibitor; - As it has a similar shape to {pyrophosphate/ substrate} / phosphate has a complementary shape the active site; - Phosphate {binds to/ competes for} the active site; - Prevents pyrophosphate binding / fewer enzyme-substrate complexes are formed; - The maximum rate of reaction can be reached at higher concentrations of pyrophosphate;						

			7-9 marks Indicative content of this level is: Detailed explanation of lock and key/ induced fit theory. Detailed explanation of the action of inhibition of phenylalanine. Detailed explanation of the action of inhibition of phosphate. <i>The candidate constructs an articulate, integrated account, correctly linking relevant points, such as those in the indicative content, which shows sequential reasoning. The answer fully addresses the question with no irrelevant inclusions or significant omissions. The candidate uses scientific conventions and vocabulary appropriately and accurately.</i>						
			4-6 marks Indicative content of this level: Any two from: Explanation of lock and key/ induced fit theory/ or description of. explanation of the action of inhibition of phenylalanine explanation of the action of inhibition of phosphate. <i>The candidate constructs an account correctly linking some relevant points, such as those in the indicative content, showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately.</i>						

			1-3 marks Indicative content of this level is: Brief explanation of lock and key/ induced fit theory OR Brief explanation of the action of inhibition of phenylalanine OR Brief explanation of the action of inhibition of phosphate. <i>The candidate makes some relevant points, such as those in the indicative content, showing limited reasoning. The answer addresses the question with significant omissions. The candidate has limited use of scientific conventions and vocabulary.</i>						
			0 marks <i>The candidate does not make any attempt or give a relevant answer worthy of credit.</i>						
			Question 6 total	4	3	2	9		